IE2024 - Probability & Statistics

Statistical Inference

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MSc, MBA, PGDip, BSc (1st Class)



Estimation of Population Parameters

Suppose that the population has an unknown parameter such as mean, variance or the proportion of success. Then an estimate of the unknown parameter can be made from the information supplied by a random sample or samples taken from the population.

A statistic is used to estimate the value of a parameter is called “Estimator” and denoted using capital letters with hat notations (Ex- $\hat{S}$). The numerical value taken by the estimator is called the estimate.

There are two ways of estimation.

1. Point estimation – Determine an estimator which based on the sample data and will give a single value estimate for an unknown parameter.
2. Interval estimation – Determine an interval that is likely to contain an unknown parameter value.

Unbiased & Most Efficient Estimators for Population Parameters

a) Population mean (μ)

$$\hat{\mu} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

a) Population variance (σ^2)

$$\widehat{\sigma^2} = S^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{X})^2$$

a) Population proportion (P)

$$\hat{P} = \frac{Number\ of\ successes\ in\ the\ sample}{n}$$

Interval Estimation

Let L and U be the lower and upper limits for a parameter θ based on a random sample $x_1, x_2, x_3, \ldots, x_n$. Both L and U are functions of the sample.

We can write the interval estimate of θ as:

$$P(L \leq \theta \leq U) = 1 - \alpha$$

- It is $(1-\alpha)100\%$ confident that the true parameter θ is located in the interval (L, U).
- (L, U) is referred to as a $(1-\alpha)100\%$ confidence interval for θ.
- In simple words, **when we create infinite intervals with the $(1-\alpha)100\%$ confidence by getting random samples of the population repeatedly, $(1-\alpha)100\%$ of them are containing the true parameter θ.**

Confidence Intervals

- A confidence interval (CI) is a range of values, derived from a data sample, that is likely to contain the true population parameter (such as a mean, proportion, or variance) with a certain degree of confidence.
- In simpler terms, it's a way of estimating an unknown value (like the average height of people in a city) by providing a range that the true value is likely to fall within.
- Suppose you calculate a 95% confidence interval for the average height of students as (165 cm, 175 cm).
    - **Interpretation**: "We are 95% confident that the true average height of all students lies between 165 cm and 175 cm.“
    - **Misconception**: It does not mean that there’s a 95% chance the true mean is in that range. Instead, if we were to repeat this sampling process many times, 95% of those intervals would contain the true population mean.

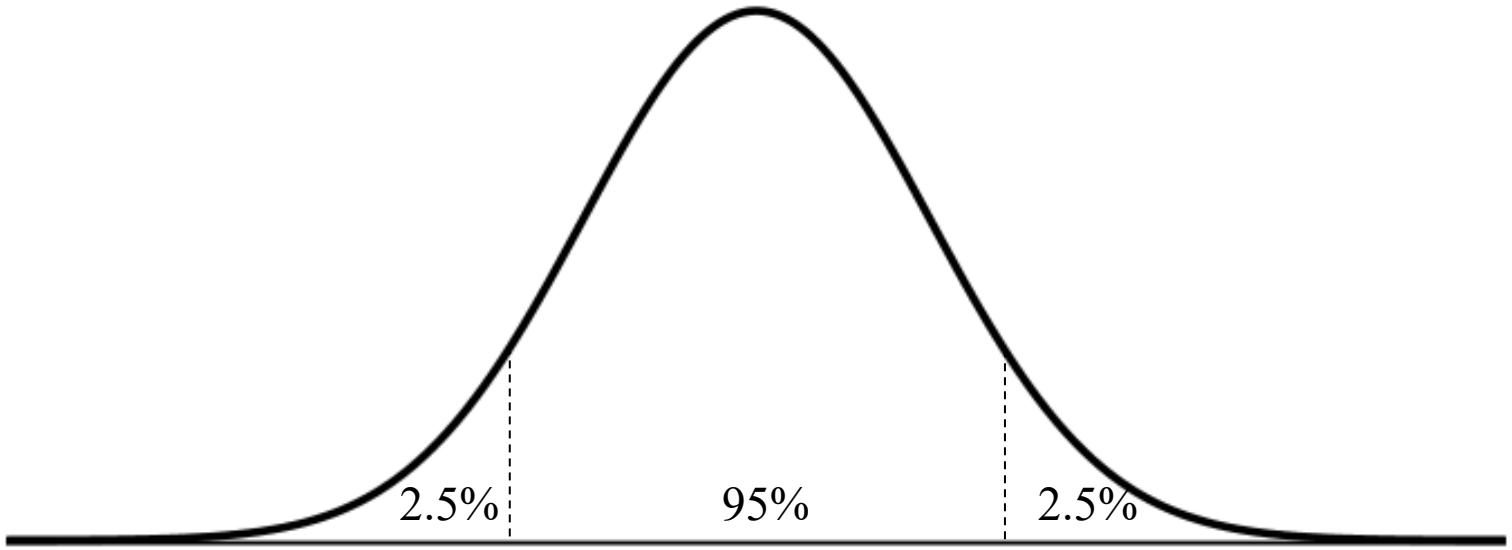
$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is known

$(1-\alpha)100\%$ confidence interval for μ is:

$$\left[\bar{X} - {}^{\sigma}/_{\sqrt{n}} Z_{\left(1-\frac{\alpha}{2}\right)} \quad , \quad \bar{X} + {}^{\sigma}/_{\sqrt{n}} Z_{(1-\frac{\alpha}{2})}\right]$$

Ex-

A random sample of 12 items is selected from a normal population with variance 4. Sample mean is 10.4. Find the 95% confidence interval for the population mean.



$$Z_{2.5\%} = -1.96 \qquad Z_{97.5\%} = 1.96$$

$$\left[10.4 - {}^{2}/_{\sqrt{12}} Z_{2.5\%} \quad , 10.4 + {}^{2}/_{\sqrt{12}} Z_{2.5\%}\right] = [9.26\,, 11.52]$$

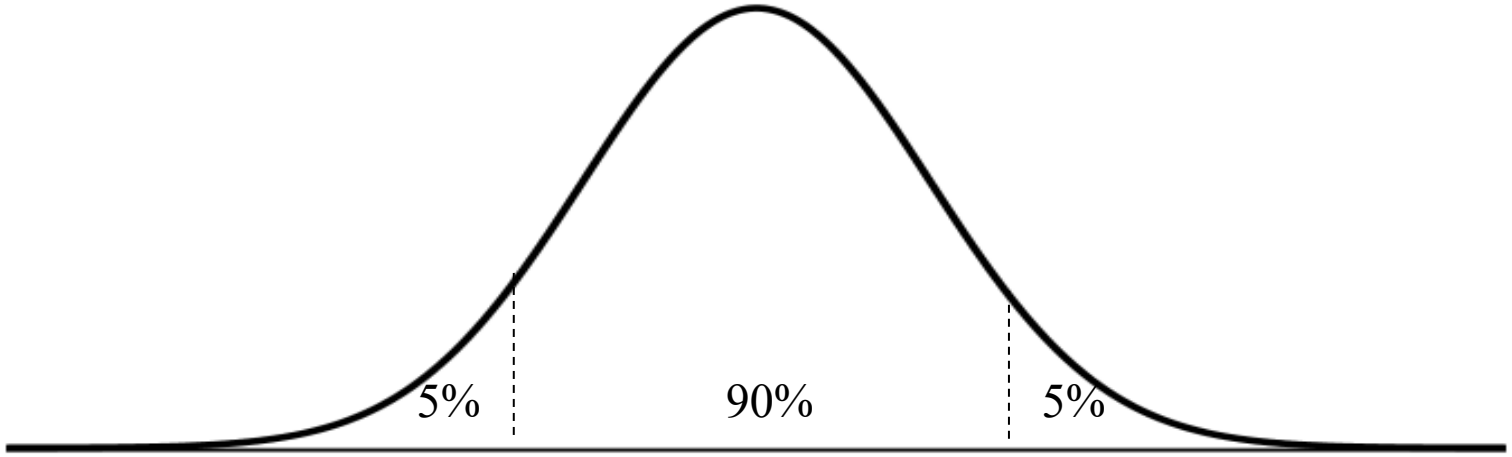
$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is unknown

$(1-\alpha)100\%$ confidence interval for μ is:

$$\left[\bar{X} - {}^{S}/_{\sqrt{n}}\, t_{(n-1),\frac{\alpha}{2}} \quad , \quad \bar{X} + {}^{S}/_{\sqrt{n}}\, t_{(n-1),\frac{\alpha}{2}}\right]$$

Ex-

Let X equal the amount of butterfat in pounds produced by a typical cow during a 105-day milk production period between her first and second calves. Assume that the distribution of X is normal but the population variance is unknown. Construct a 90% confidence interval for population mean when the sample mean is 507.7 and sample standard deviation is 89.75. (n=20)



$$-t_{(20-1),5\%} = -1.729 \qquad t_{(20-1),5\%} = 1.729$$

$$\left[507.5 - {}^{89.75}/_{\sqrt{20}}\, t_{(20-1),5\%},\ 10.4 + {}^{89.75}/_{\sqrt{20}}\, t_{(20-1),5\%}\right] = [472.801\ ,542.199]$$

Hypothesis Testing

When making a statistical inquiry, we often put forward a claim (hypothesis) about a population parameter.

Ex-

- Mean height of 15 year old boys is 1.62m
- The proportion of diabetes in a particular district is 0.05
- The mean number of telephone calls received by a telephone operator over a period of 60 minutes is 20.



Hypothesis Testing

The hypothesis we put forward is called the **Null Hypothesis** and is denoted by $\boldsymbol{H_0}$. Just define Null Hypotheses for above examples as below.

- $H_0 : \mu = 1.62$
- $H_0 : p = 0.05$
- $H_0 : \mu = 20$

In order to test the validity of H_0 we consider observations made on the random sample taken from the populations and perform a statistical test.

Hypothesis Testing

This test shows that whether we should reject the null hypothesis (rejecting the claim stated under H_0), we do so in favor of an **Alternative Hypothesis**, which is usually the opposite of what is stated in the null hypothesis and is denoted by $\boldsymbol{H_1}$ or $\boldsymbol{H_a}$.

Ex-

Took a sample (income) and found that $\bar{X}$=40000. Suppose we want to test the hypothesis that the mean income is greater than 37000. Then,

$$H_0 : \mu = 37000 \; and \; H_1 : \mu > 37000$$

Defining Hypotheses

Putting the question under the alternative hypothesis.

Ex-

1. The UGC Sri Lanka needs to know that whether the “majority of students at University of Moratuwa are from Western Province”

$$\boldsymbol{H_1 : p > 0.5} \quad \textit{and} \quad \boldsymbol{H_0 : p \leq 0.5} \longrightarrow \boldsymbol{H_0 : p = 0.5}$$

1. A consumer test agency wants to see whether the mean lifetime of a brand of tires is less than 42000 miles as the tire manufacturer advertises that the average lifetime is at least 42000.
2. The length of a certain lumber from a national home building store is supposed to be 8.5 feet. A builder wants to check whether the shipment of lumber she receives has a mean length different from 8.5 feet.
3. A political news company believes the national approval rating for the current president has fallen below 40%.

Defining Hypotheses

2. A consumer test agency wants to see whether the mean lifetime of a brand of tires is less than 42000 miles as the tire manufacturer advertises that the average lifetime is at least 42000.

$$H_1 : \mu < 42000 \quad and \quad H_0 : \mu = 42000$$

3. The length of a certain lumber from a national home building store is supposed to be 8.5 feet. A builder wants to check whether the shipment of lumber she receives has a mean length different from 8.5 feet.

$$H_1 : \mu \neq 8.5 \quad and \quad H_0 : \mu = 8.5$$

4. A political news company believes the national approval rating for the current president has fallen below 40%.

$$H_1 : p < 0.4 \quad and \quad H_0 : p = 0.4$$

Type I Versus Type II Errors

When testing hypotheses there are essential two errors that can be made.

1. **Type I error** - Rejecting the null hypothesis when the null is correct actually.
2. **Type II error** - Not rejecting the null hypothesis when the null is not correct actually.

Decision	Reality	
	H_0 is true	H_0 is false
Reject H_0	Type I error	Correct
Do not reject H_0	Correct	Type II error

$$P(Type\ I\ error) = \alpha$$

$$P(Type\ II\ error) = \beta$$

$$Power = 1 - \beta$$

Significance Level $P(Type\ I\ error)$

When we conduct a hypothesis test we will always have to specify a significance level. We basically have to answer the question – “What is the probability of a Type I error we are willing to live with?”.

We cannot generally quantify the probability of a type II error. To avoid making Type ii errors we will always say “Do not reject the null hypothesis” instead of “Accept the null hypothesis”.



Popular Significance Levels

0.1 0.05 0.01

Perform Hypothesis Testing

Classical Approach:

- Define null and alternative hypothesis.
- Check whether the hypothesis is one tailed or two tailed.
- Find the appropriate test statistic and its distribution.
- Compute the test statistic under H_0.
- Define the rejection region according to the given significance level.
- If the calculated test statistic falls in the rejection region, reject H_0 and if not, do not reject H_0.

Tests About Mean : Z – test when population variance is known

Consider the following hypothesis for the population mean, suppose to be tested using a random sample of size n. **This is a two tailed test**.

$$H_0 : \mu = A$$

$$H_1 : \mu \neq A$$

Get a sample from the population and calculate the sample mean $\bar{X}$.

Now consider the test statistic,

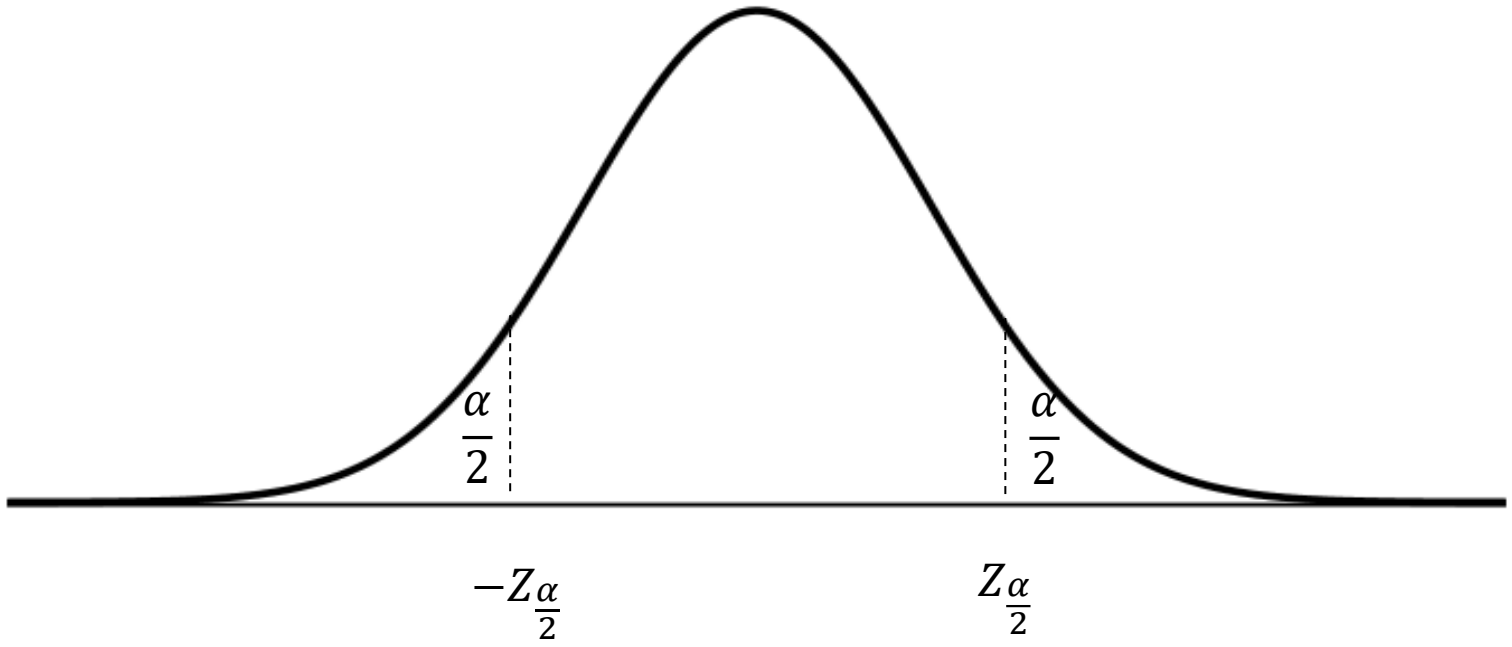
$$\mathrm{Z} = \frac{(\bar{X} - \mu)}{\sigma / \sqrt{n}} \sim N(0,1)$$

Now compute the values of the test statistic under H_0. Then,

$$\mathrm{Z}_{cal} = \frac{(\bar{X} - A)}{\sigma / \sqrt{n}}$$

Tests About Mean : Z – test when population variance is known

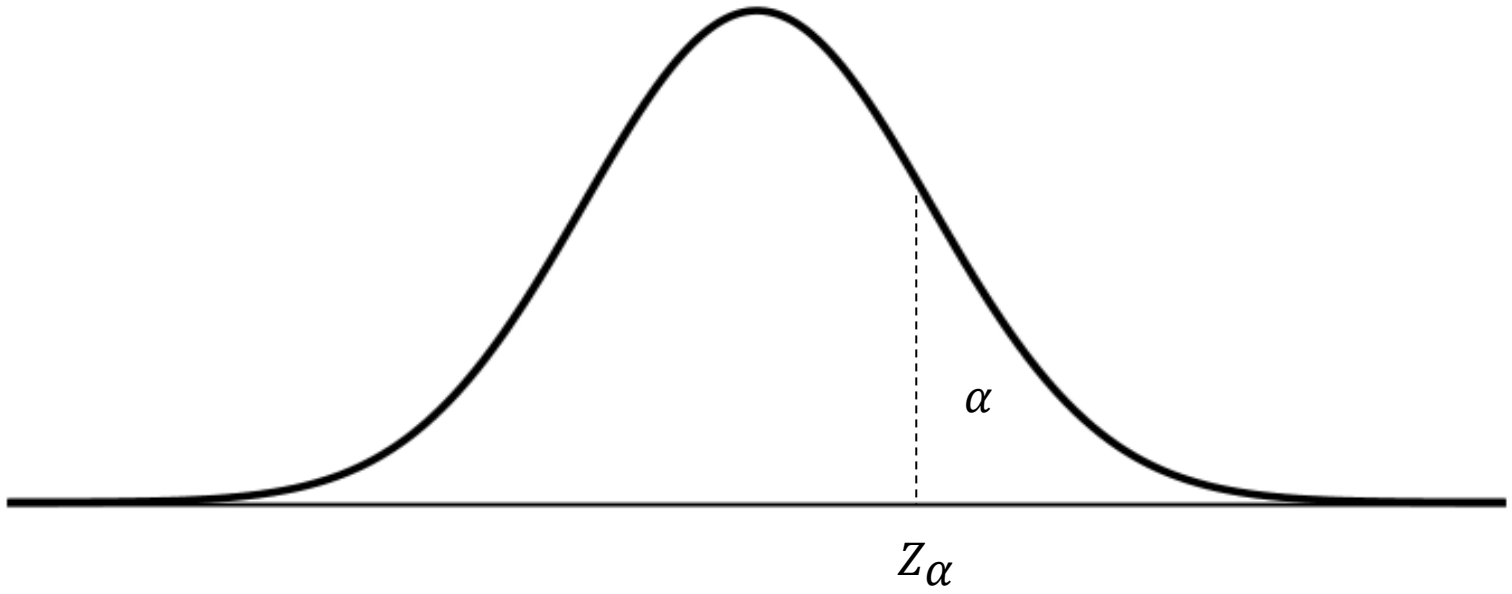
Since this is a two tail test,



If $Z_{cal} > Z_{\frac{\alpha}{2}}$ or $Z_{cal} < -Z_{\frac{\alpha}{2}}$ we reject H_0

Tests About Mean : Z – test when population variance is known

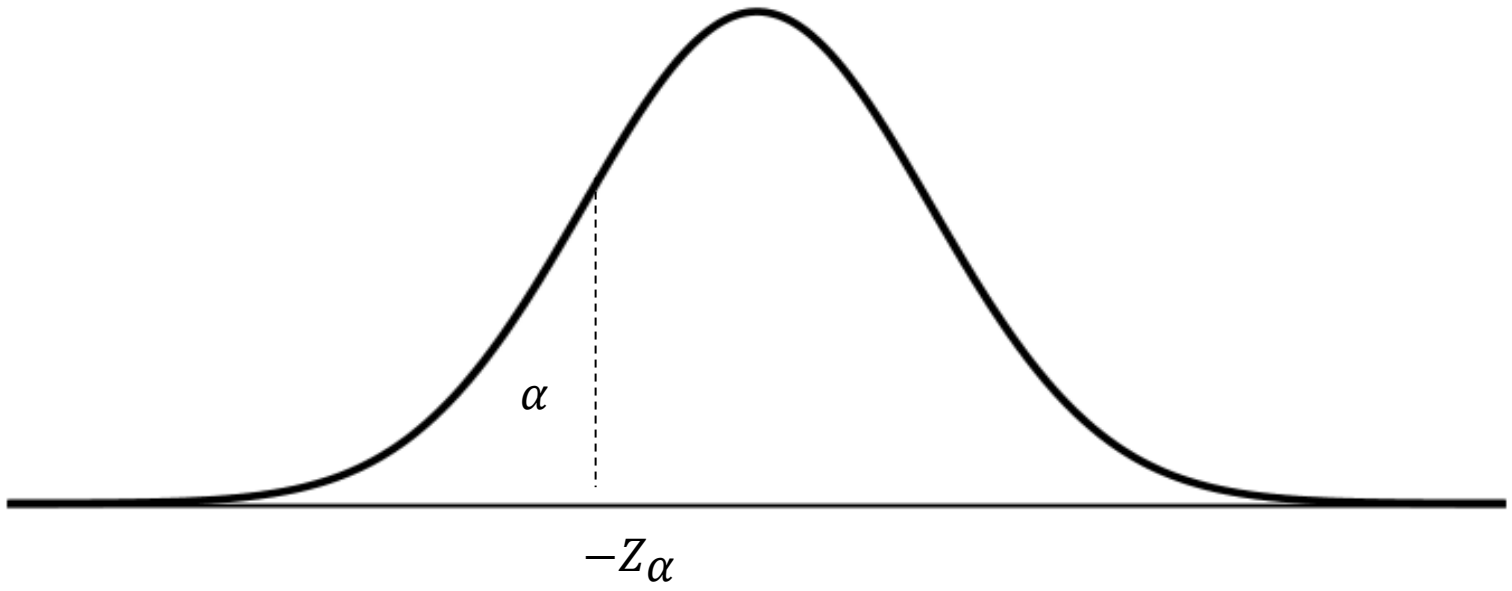
In a one tail – right tail test,, $H_0 : \mu = A$ and $H_1 : \mu > A$



If $Z_{cal} > Z_\alpha$ we reject H_0

Tests About Mean : Z – test when population variance is known

In a one tail – left tail test,, $H_0 : \mu = A$ and $H_1 : \mu < A$



If $Z_{cal} < -Z_\alpha$ we reject H_0

Ex-

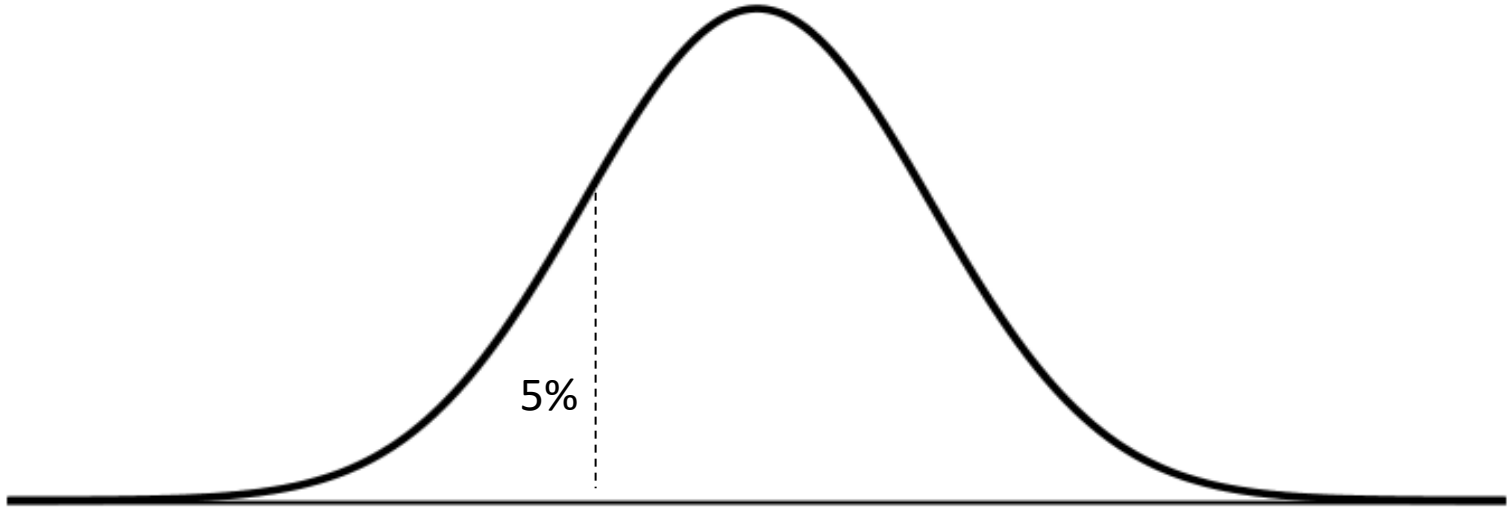
Boys of a certain age are known to have a mean weight of 85 pounds. A complaint is made that the boys living in a municipal children's home are underfed. 25 boys from the home are weighed and found to have a mean weight of 80.94 pounds. It is known that the population standard deviation is 11.6 pounds. Based on the available data what should be concluded concerning the complaint? (Significance level 5%)

$$H_0 : \mu = 85$$

$$H_1 : \mu < 85$$

Under H_0 ,

$$\mathrm{Z}_{cal} = \frac{(80.94 - 85)}{11.6/\sqrt{25}} = -1.75$$



$-Z_{5\%} = -1.645$

$Z_{cal} = -1.75 < -Z_{5\%}$,so we reject H_0 at 5% level of significance.

Conclusion **– There is an evidence to say that, the boys living in a municipal children's home are underfed at 5% level of significance.**

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Tests About Mean : t – test when population variance is unknown

Consider the following hypothesis for the population mean, suppose to be tested using a random sample of size n. **This is a two tailed test**.

$$H_0 : \mu = A$$

$$H_1 : \mu \neq A$$

Get a sample from the population and calculate the sample mean $\bar{X}$.

Now consider the test statistic,

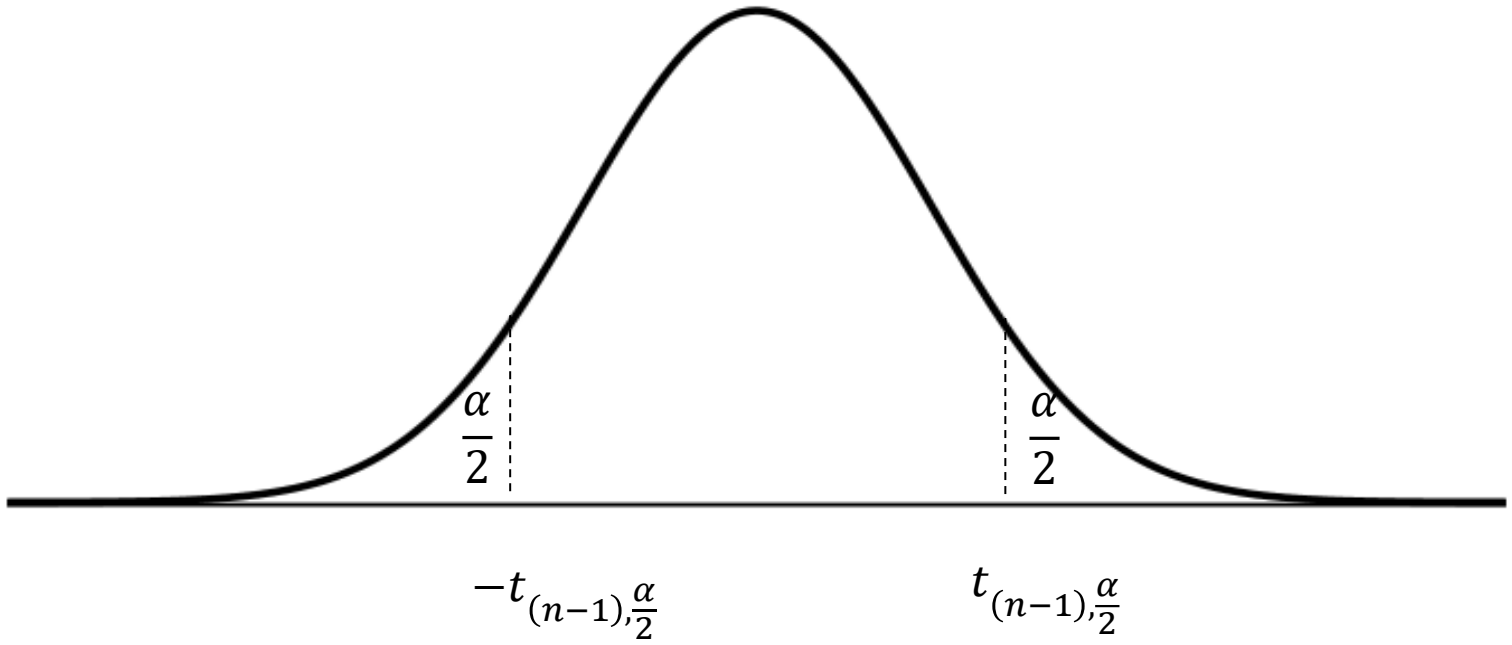
$$\mathrm{T} = \frac{(\bar{X} - \mu)}{S/\sqrt{n}} \sim t_{(n-1)}$$

Now compute the values of the test statistic under H_0. Then,

$$\mathrm{t}_{cal} = \frac{(\bar{X} - A)}{S/\sqrt{n}}$$

Tests About Mean : t – test when population variance is unknown

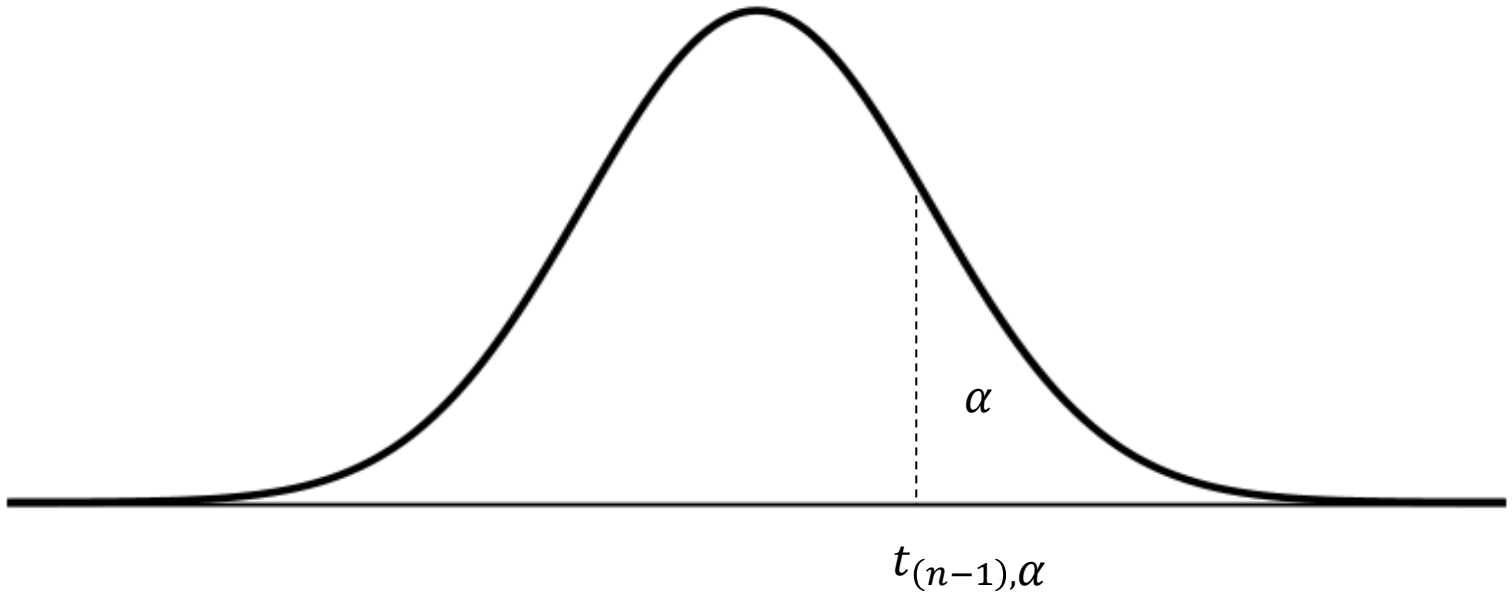
Since this is a two tail test,



If $t_{cal} > t_{(n-1),\frac{\alpha}{2}}$ or $t_{cal} < -t_{(n-1),\frac{\alpha}{2}}$ we reject H_0

Tests About Mean : t – test when population variance is unknown

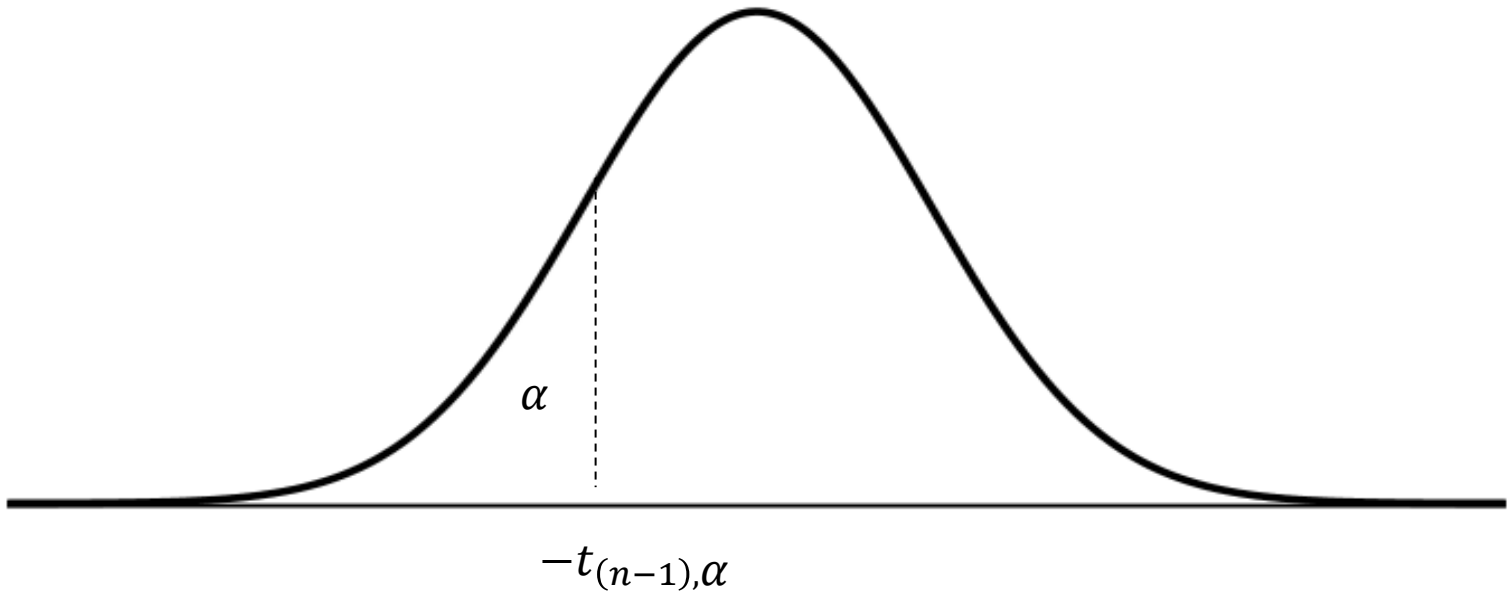
In a one tail – right tail test,, $H_0 : \mu = A$ and $H_1 : \mu > A$



If $t_{cal} > t_{(n-1),\alpha}$ we reject H_0

Tests About Mean : t – test when population variance is unknown

In a one tail – left tail test,, $H_0 : \mu = A$ and $H_1 : \mu < A$



If $t_{cal} < -t_{(n-1),\alpha}$ we reject H_0

Ex-

Five readings of the resistance in Ohms (Ω) of a piece of wire gave the following results.

$$1.15, 1.49, 1.54, 1.52, 1.54$$

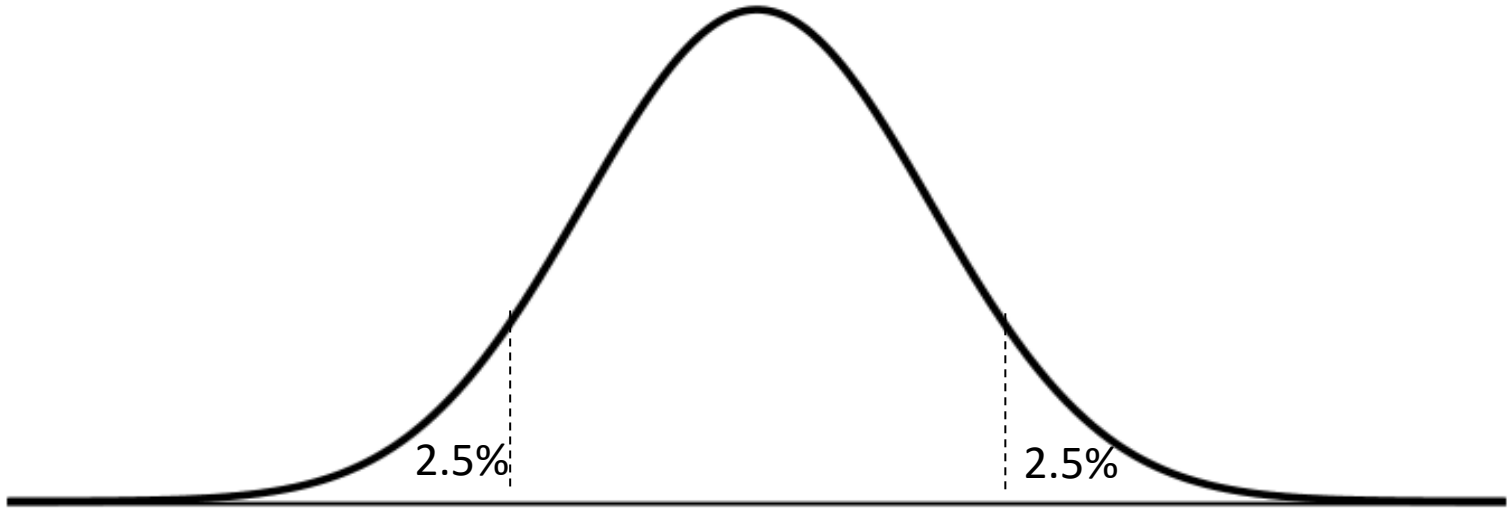
If the wire is pure silver, its resistance should be 1.5Ω. Test at 5% level of significance whether the wire is pure silver. Assume the normality of the readings.

$$H_0 : \mu = 1.5$$

$$H_1 : \mu \neq 1.5$$

Under H_0 ,

$$\mathrm{t}_{cal} = \frac{(1.448 - 1.5)}{0.15/\sqrt{5}} = -0.78$$



$-t_{(4),2.5\%} = -2.7763 \qquad t_{(4),2.5\%} = 2.7763$

$\mathrm{t}_{cal} = -0.78 > -t_{(4),2.5\%}$,so we do not reject H_0 at 5% level of significance.

Conclusion - **There is an evidence to say that the wire is pure silver at 5% level of significance.**

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

Two Population Means

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Case A - $\sigma_1^2 \neq \sigma_2^2$ but known

$$(\bar{X} - \bar{Y}) \pm Z_{\frac{\alpha}{2}} \sqrt{{\sigma_1^2}/{n} + {\sigma_2^2}/{m}}$$

Case B - $\sigma_1^2 = \sigma_2^2$ but known

$$(\bar{X} - \bar{Y}) \pm Z_{\frac{\alpha}{2}} \left(\sigma \sqrt{{1}/{n} + {1}/{m}} \right)$$

(1−α)100% Confidence Interval for the difference between two normally distributed population means.

Case C - $\sigma_1^2 = \sigma_2^2$ but unknown

So the (1−α)100% Confidence Interval for $\mu_1 - \mu_2$ is,

$$(\bar{X} - \bar{Y}) \pm t_{(n+m-2),\frac{\alpha}{2}} \; S_p \sqrt{1/n + 1/m}$$

If n and m are very large (say 30), the (1−α)100% Confidence Interval for $\mu_1 - \mu_2$ can be written as,

$$(\bar{X} - \bar{Y}) \pm Z_{\frac{\alpha}{2}} \, S_p \sqrt{1/n + 1/m}$$

$$S_p = \sqrt{\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$$

(1−α)100% Confidence Interval for the difference between two normally distributed population means.

Case D - $\sigma_1^2 \neq \sigma_2^2$ but unknown

The (1−α)100% Confidence Interval for $\mu_1 - \mu_2$ is,

$$(\bar{X} - \bar{Y}) \pm t_{(df),\frac{\alpha}{2}} \sqrt{S_1^2/n + S_2^2/m}$$

$$df = \frac{\left(S_1^2/n + S_2^2/m\right)^2}{\frac{\left(S_1^2/n\right)^2}{n-1} + \frac{\left(S_2^2/m\right)^2}{m-1}}$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Ex-

From a large class 4 students were drawn and their Maths marks were 64, 66, 89 and 77. From a second large class 3 students were drawn and their Maths marks were 56, 71 and 53. Assuming the classes are normally distributed and have equal variances, find 95% confidence interval for the difference between the mean of the two Maths marks.

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Ex-

From a large class 4 students were drawn and their Maths marks were 64, 66, 89 and 77. From a second large class 3 students were drawn and their Maths marks were 56, 71 and 53. Assuming the classes are normally distributed and have equal variances, find 95% confidence interval for the difference between the mean of the two Maths marks.

$$\bar{X} = 74 \; and \bar{Y} = 60$$

$$S_1^2 = 132.67 \text{ and } S_2^2 = 92$$

$$S_p = \sqrt{\frac{3 \times 132.67 + 2 \times 92}{4 + 3 - 2}} = 10.78$$

$$t_{2.5\%,5} = 2.57$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

$$(\bar{X}-\bar{Y}) \pm t_{(n+m-2),\frac{\alpha}{2}}\ S_p\sqrt{1/n + 1/m}$$

$$14 \pm 10.78\sqrt{\frac{1}{4}+\frac{1}{3}} \times 2.57$$

$$[-7.034, 35.09]$$

Comparing two population means – Unpaired case

Consider following hypotheses,

$$H_0 : \mu_1 - \mu_2 = 0 \, , \, H_1 : \mu_1 - \mu_2 \neq 0$$

$$H_0 : \mu_1 - \mu_2 = 0 \, , \, H_1 : \mu_1 - \mu_2 > 0$$

$$H_0 : \mu_1 - \mu_2 = 0 \, , \, H_1 : \mu_1 - \mu_2 < 0$$

Comparing two population means – Unpaired case.

Case A - $\sigma_1^2 \neq \sigma_2^2$ but known

Then the test statistic is,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \sim N(0,1)$$

Case B - $\sigma_1^2 = \sigma_2^2$ but known

Then the test statistic is,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n + 1/m}} \sim N(0,1)$$

Comparing two population means – Unpaired case.

Case C - $\sigma_1^2 = \sigma_2^2$ but unknown

Then the test statistic is,

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p\sqrt{1/n + 1/m}} \sim t_{(n+m-2)}$$

Where,

$$S_p = \sqrt{\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$$

Comparing two population means – Unpaired case.

Case D - $\sigma_1^2 \neq \sigma_2^2$ but unknown

Then the test statistic is,

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n + S_2^2/m}} \sim t_{(df)}$$

Where,

$$df = \frac{\left(S_1^2/n + S_2^2/m\right)^2}{\frac{\left(S_1^2/n\right)^2}{n-1} + \frac{\left(S_2^2/m\right)^2}{m-1}}$$

Comparing two population means – Unpaired case.

Ex-

A test was given to a group of 100 scouts and 144 guides. The mean score for scouts was 27.53 and the mean score for the guides was 26.81. Assuming a common population standard deviation of 3.48, test at 5% level of significance whether the scouts performance in the test was better than that of the guides. Assume that the scores are normally distributed.

Comparing two population means – Unpaired case.

Ex-

A test was given to a group of 100 scouts and 144 guides. The mean score for scouts was 27.53 and the mean score for the guides was 26.81. Assuming a common population standard deviation of 3.48, test at 5% level of significance whether the scouts performance in the test was better than that of the guides. Assume that the scores are normally distributed.

$$n_1 = 100 \qquad n_2 = 144$$

$$\bar{X} = 27.53 \qquad \bar{Y} = 26.81$$

$$\sigma = 3.48$$

$$H_0 : \mu_1 - \mu_2 = 0 \text{ , } H_1 : \mu_1 - \mu_2 > 0$$

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n + 1/m}} \sim N(0,1)$$

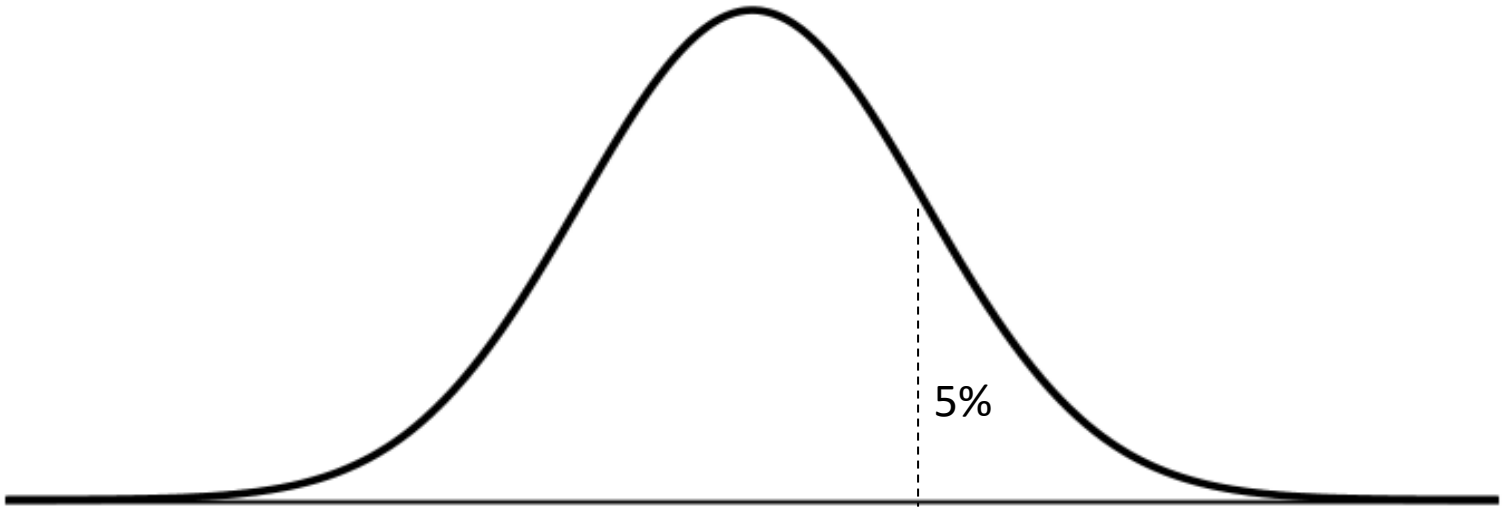
Comparing two population means – Unpaired case.

Under H_0,

$$Z_{cal} = \frac{(27.53 - 26.81) - (0)}{3.41\sqrt{{}^{1}/_{100} + {}^{1}/_{144}}} = 1.5894$$

This is one tail right tail test.

$Z_{cal} < Z_{critical}$



$Z_{critical} = 1.645$

Comparing two population means – Unpaired case.

Do not reject H_0 5% level of significance.

Conclusion - **There is no evidence to say that the scouts performance in the test was better than that of the guides at 5% level of significance.**

Comparing two population means – Paired case.

The objective of this section is testing the difference between population means based on means of two dependent samples.

Paired samples: The sample selected from the first population is related to the corresponding sample from the second population.

Assumptions we made under paired samples:

The difference of the pairs follow a normal distribution or the number of pairs is large. (note here that if the number of pairs is less than 30, we need to check whether the **differences** are normally distributed, but we do not need to check for the normality of each population.)

The key to analyze a matched sample design is to use the difference data.

Comparing two population means – Paired case.

Consider hypothesis,

$$H_0 : \mu_1 - \mu_2 = 0 \, , H_1 : \mu_1 - \mu_2 \neq 0$$

$$H_0 : \mu_1 - \mu_2 = 0 \, , H_1 : \mu_1 - \mu_2 > 0$$

$$H_0 : \mu_1 - \mu_2 = 0 \, , H_1 : \mu_1 - \mu_2 < 0$$

Test statistic:

Let d=differences between the pairs of data, then $\bar{d}$=mean of these differences. Then the test statistic is,

$$T = \frac{\bar{d} - \mu_{def}}{S_{def} \Big/ \sqrt{n}} \sim t_{(n-1)}$$

$$\mu_{def = \mu_1 - \mu_2}$$

Where n denotes the number of pairs or the number of differences.

Comparing two population means – Paired case.

Ex-

Trace metals in drinking water affect the flavor and unusually high concentration can pose a health hazard. Ten pairs of data were taken measuring zinc concentration in bottom water and surface water. Does the data suggest that the true average concentration in the bottom water exceeds that of surface water? (5% level of significance)

	Location									
	1	2	3	4	5	6	7	8	9	10
Zinc concentration in bottom water	0.430	0.266	0.567	0.531	0.707	0.716	0.651	0.589	0.469	0.723
Zinc concentration in surface water	0.415	0.238	0.390	0.410	0.605	0.609	0.632	0.523	0.411	0.612

Comparing two population means – Paired case.

Since the sample size is not large enough (Less than 30), we need to check whether the differences follow a normal distribution.

	B	S	D
0	0.430	0.415	0.015
1	0.266	0.238	0.028
2	0.567	0.390	0.177
3	0.531	0.410	0.121
4	0.707	0.605	0.102
5	0.716	0.609	0.107
6	0.651	0.632	0.019
7	0.589	0.523	0.066
8	0.469	0.411	0.058
9	0.723	0.612	0.111

Thus, we can test and show that the difference may come from a normal distribution

Comparing two population means – Paired case.

$$H_0 : \mu_{def} = 0 \, , \, H_1 : \mu_{def} > 0$$

$$T_{cal} = 4.86$$

$$t_{0.05,9} = 1.833$$

Since $T_{cal} > t_{0.05,9}$, Reject H_0 at 5% level of significance.

Conclusion - **There is an evidence to say that the true average concentration in the bottom water exceeds that of surface water at 5% level of significance.**

Analysis of Variance (ANOVA)

Comparing Multiple Population Means

Analysis of Variance (ANOVA)

An extension of the pooled variance t-test for the difference between two means. Though the term mentions the word variance, we are testing differences in means. ANOVA separates total variation into variation within (random variation) and among the groups. So long as the ratio between within and among variation is large enough, we conclude that there are significant differences between groups.

Assumptions:

- Populations are normally distributed.
- Populations have equal variances.
- Samples are randomly and independently drawn.

One Way ANOVA

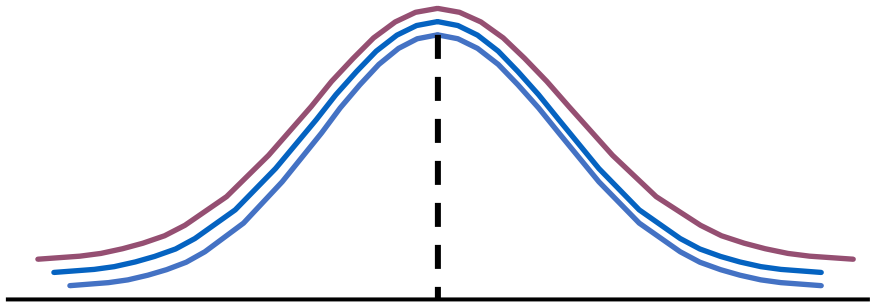
$$H_0: \mu_1 = \mu_2 = \cdots = \mu_c$$

All population means are equal. (no variation in means among groups)

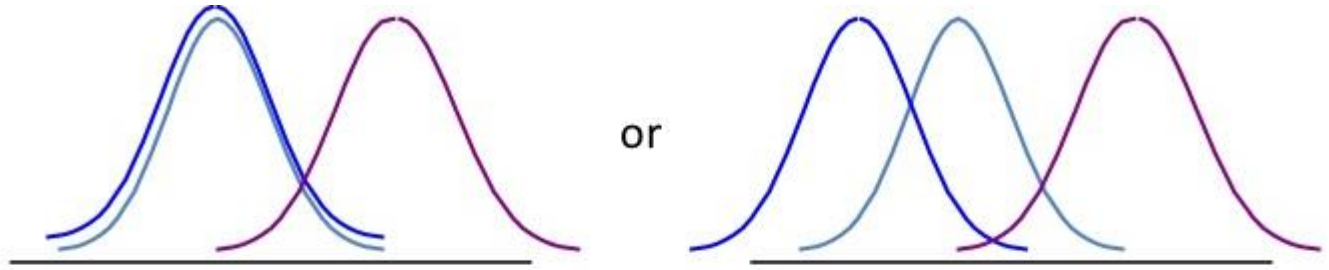
$$H_1: At\ least\ one\ population\ mean\ is\ different$$

At least one population mean is different. (not necessarily that all population means are different)

H_0

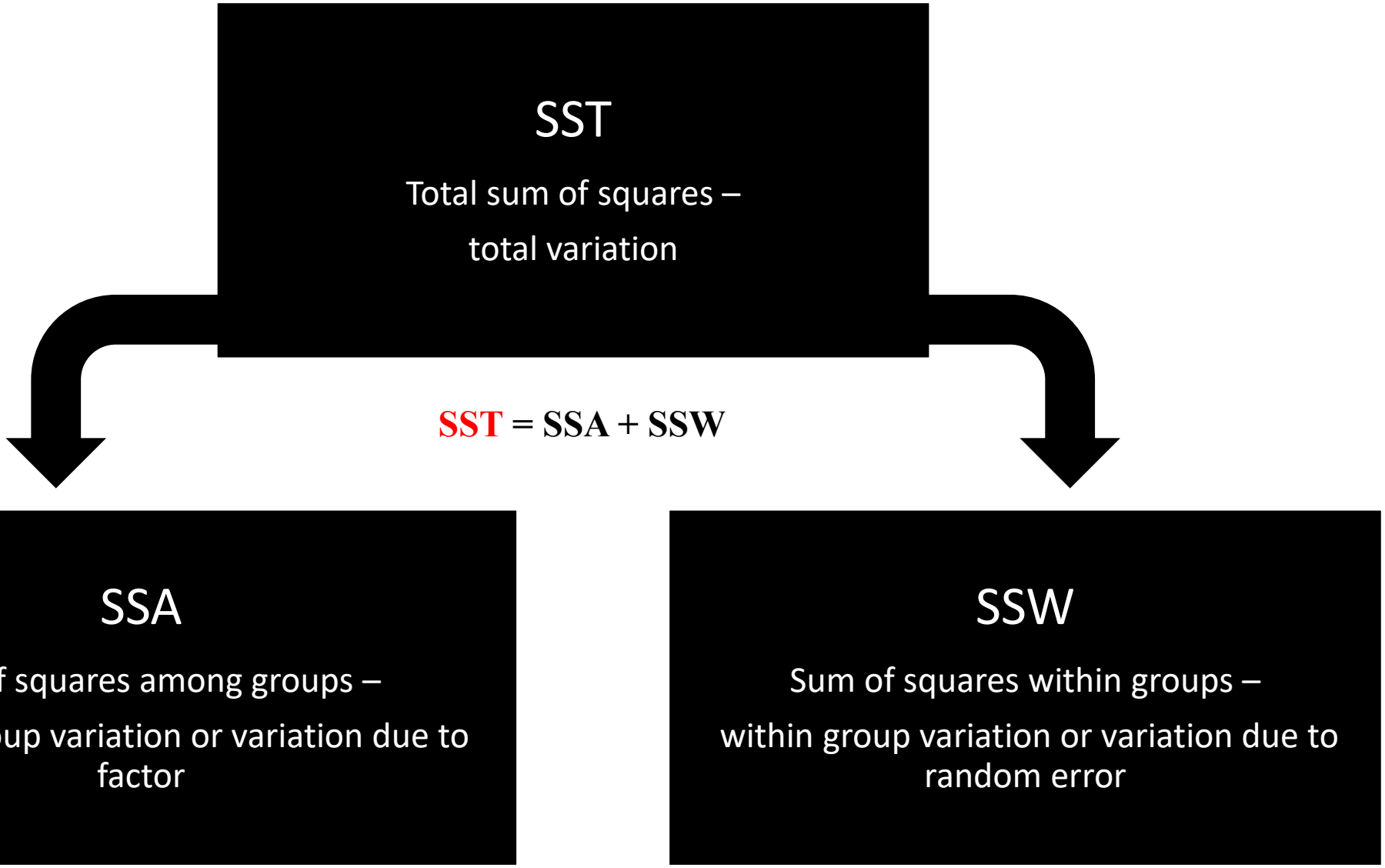


H_1



$\mu_1 = \mu_2 \neq \mu_3$ $\mu_1 \neq \mu_2 \neq \mu_3$

One Way ANOVA



One Way ANOVA

$$SST = \sum_{j=1}^{c} \sum_{i=1}^{n_j} \left(X_{ij} - \bar{\bar{X}}\right)^2$$

- c = number of groups or levels
- n_j = number of observations in group j
- X_{ij} = i^{th} observation from group j
- $\bar{\bar{X}}$ = grand mean (mean of all data values)

One Way ANOVA

$$SSA = \sum_{j=1}^{c} n_j \left(\bar{X}_j - \bar{\bar{X}}\right)^2$$

- $\bar{X}_j$ = sample mean from group j

$$Mean\ Square\ Among\ (MSA) = \frac{SSA}{c-1}$$

- where c-1 is the degrees of freedom of the SSA

One Way ANOVA

$$SSW = \sum_{j=1}^{c} \sum_{i=i}^{n_j} \left(X_{ij} - \bar{X}_j\right)^2$$

- $\bar{X}_j$ = sample mean from group j

$$Mean\ Square\ Within\ (MSW) = \frac{SSW}{n - c}$$

- Where n-c is the degrees of freedom of the SSW

One Way ANOVA Table

Source of variation	Degrees of freedom	Mean square	F
Between groups	c-1	$MSA = \frac{SSA}{c-1}$	$F_{stat} = \frac{MSA}{MSW}$
Within groups	n-c	$MSW = \frac{SSW}{n-c}$	
Total	n-c+(c-1)=n-1	$MST = \frac{SST}{n-1}$	

One Way ANOVA Test

The test statistic is the ratio of the mean of among variance and mean of within variance. The ratio must always be positive.

- c-1 is typically small
- n-c is typically large

Decision rule: Reject H_0 if $F_{stat} > F_{\alpha}$. Otherwise, do not reject.

One Way ANOVA Test

Ex:-

You want to see if three different tennis rackets hit the ball different distances. You randomly select five measurements from trials on an automated ball machine. At the 5% significance level, is there a difference in mean distance?

Racket 1	Racket 2	Racket 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

One Way ANOVA Test

Ex:-

When the needful calculations are done,

$$\bar{\bar{X}} = 227 \quad \bar{X}_1 = 249.2 \quad \bar{X}_2 = 226 \quad \bar{X}_3 = 205.8 \quad n_1 = n_2 = n_3 = 5 \quad n = 15 \quad c = 3$$

$$SSA = 4{,}716.4$$
$$SSW = 1{,}119.6$$

$$MSA = 4716.4/(3-1) = 2{,}358.2$$
$$MSW = 1119.6/(15-3) = 93.3$$

$$F_{stat} = \frac{2358.2}{93.3} = 25.275, \qquad F_{0.05,2,12} = 3.89$$

Since $F_{stat} > F_{0.05,2,12}$, Reject H_0 at 5% level of significance.

Conclusion: There is an evidence to say that at least one mean is different from the rest at 5% level of significance.

Post – Hoc Analysis : Mean Comparisons

When the null hypothesis of ANOVA is rejected, it only suggests that there are significant differences between means. However, it does not specify which of the treatment means differ. Therefore further comparisons between group or treatment is required.

- Tukey's Honestly Significant Difference (HSD)
- Bonferroni Correction
- Scheffé's Method
- Dunnett's Test (vs. a control group)
- Fisher's Least Significant Difference (LSD)
- Newman-Keuls Method
- Holm's Sequential Bonferroni
- Hochberg's GT2
- Gabriel's Procedure
- Games-Howell Test (for unequal variances and sample sizes)



Chi Square Test

Categorical Data Analysis

Chi Square Test for Two Proportions

We now consider an alternative testing procedure that can be used when comparing counts of categorical responses (comparing proportions) between two independent groups. We start by developing contingency tables.

	COLUMN VARIABLE (GROUP)		
ROW VARIABLE	1	2	Totals
Items of interest	X_1	X_2	X
Items not of interest	$n_1 - X_1$	$n_2 - X_2$	$n - X$
Totals	n_1	n_2	n

Chi Square Test for Two Proportions

Consider following hypothesis,

$$H_0 : p_1 = p_2 \text{ , } H_1 : p_1 \neq p_2$$

Then the test statistic is,

$$\chi^2_{stat} = \sum_{all\ cells} \frac{(f_o - f_e)^2}{f_e} \sim \chi^2_{(1)}$$

Chi Square Test for Two Proportions

where f_o is the observed frequency and f_e is the expected frequency if the null hypothesis is true and the 2×2 contingency table gives $(2-1) \times (2-1) = 1$ degree of freedom.

To compute f_e, we combine the sample proportions to estimate the common population parameter p as:

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2}$$

and separate the observations within each sample using this common proportion

The rejection rule: if $\chi^2_{stat} > \chi^2_{\alpha,1}$, reject H_0.

Important Facts About Chi Square Test

- The chi-square test with 1 degree of freedom and the Z-test are related with $\hat{Z}^2 = \chi^2_{stat}$.
- For the chi-square test to give accurate results for a 2x2 table, each expected frequency should be at least 5.
- Chi-square tests can't be used to determine evidence of a directional difference (such as $p_1 > p_2$) – for this you must use the Z-test.
- Chi-square tests are useful when you want to compare differences in proportions among more than two groups – the Z-test can't be done if there are more than 2 groups.
- The Chi-square test can be generalized to a test of independence for two categorical variables.

Chi Square Test for Two Proportions

Ex:-

A survey reports that 44% of internet users aged 18-29 have taken steps to restrict the information available about themselves online whereas only 20% of internet users aged 65+ have done the same thing. Suppose the survey consisted of 100 individuals in each age group.

At 5% significance, test that the proportion of internet users aged 18-29 who have taken steps to restrict information available about themselves online is not equal to the proportion of internet users older than 65 who have done the same thing.

$$H_0 : p_{18-29} = p_{65+}, H_1 : p_{18-29} \neq p_{65+}$$

Chi Square Test for Two Proportions

Ex:-

Let's create the contingency table.

	18-29	65+	Total
Individuals of interest	44	20	64
Individuals not of interest	56	80	136
Total	100	100	200

$$\bar{p} = \frac{44 + 20}{100 + 100} = 0.32$$

Chi Square Test for Two Proportions

Ex:-

Let's compute the test statistic.

f_o	f_e	$f_o - f_e$	$\frac{(f_o - f_e)^2}{f_e}$
44	100*0.32=32	12	144/32=4.5
20	100*0.32=32	-12	144/32=4.5
56	100*0.68=68	-12	144/68=2.1
80	100*0.68=68	12	144/68=2.1

$$\chi^2_{stat} = \sum_{all\ cells} \frac{(f_o - f_e)^2}{f_e} = 13.2$$

$$\chi^2_{5\%,1} = 3.84$$

Since $\chi^2_{stat} > \chi^2_{5\%,1}$, H_0 will be rejected at 5% level of significance.

Conclusion: There is an evidence to say that the proportion of internet users aged 18-29 who have taken steps to restrict information available about themselves online is not equal to the proportion of internet users older than 65 who have done the same thing at 5% level of significance.

Comparing Several Population Proportions

It is possible to extend the chi-squared test to the case with more than 2 independent populations.

$$H_0: p_1 = p_2 = \cdots = p_c$$
$$H_1: At\ least\ one\ of\ the\ p_j\ are\ different$$

$$\chi^2_{stat} = \sum_{all\ cells} \frac{(f_o - f_e)^2}{f_e} \sim \chi^2_{(c-1)\times(r-1)}$$

Where c and r are number of categories in each variable.

- Overall proportion is: $\bar{p} = \frac{X_1+X_2+\cdots+X_c}{n_1+n_2+\cdots+nc} = \frac{X}{n}$ where c is the number of groups.
- The rest of the test is the same as in the 2x2 case.

Assumption: Each cell in the contingency table has expected frequency of at least 1

Comparing Several Population Proportions

Ex:-
A university is thinking of switching to a trimester academic calendar. A random sample of 100 administrators, 50 students and 50 faculty members were surveyed.

Opinion	Administrators	Students	Faculty	Sum
Favor	63	20	37	120
Oppose	37	30	13	80
Total	100	50	50	200

Do some groups have a different attitude? Test at 1% level of significance.

Comparing Several Population Proportions

Ex:-

$$H_0: p_1 = p_2 = p_3 \quad \& \quad H_1: At\ least\ one\ of\ the\ p_j\ are\ different$$

$$\bar{p} = \frac{63 + 20 + 37}{100 + 50 + 50} = \frac{120}{200} = 0.6$$

f_o	f_e	$f_o - f_e$	$\frac{(f_o - f_e)^2}{f_e}$
63	100*0.6=60	3	9/60=0.15
20	50*0.6=30	-10	100/30=3.33
37	50*0.6=30	7	49/30=1.63
37	100*0.4=40	-3	9/40=0.225
30	50*0.4=20	10	100/20=5
13	50*0.4=20	-7	49/20=2.45

$$\chi^2_{stat} = \sum_{all\ cells} \frac{(f_o - f_e)^2}{f_e} = 12.8$$

$$\chi^2_{1\%,(2-1)*(3-1)} = 9.2$$

Since $\chi^2_{stat} > \chi^2_{1\%,(2-1)\times(3-1)}$, reject H_0.

Conclusion: There is an evidence to say that some groups have different attitudes at 1% level of significance.

Post – Hoc Analysis : Proportion Comparisons

Similar Mean Comparisons in ANOVA, we have the **Marascuilo Procedure** to check which groups are different. This is used when the null hypothesis of equal proportions is rejected and enables comparisons between all pairs. Similar to the Tukey procedure, start with the observed differences in proportions for all pairs of groups and then compare the absolute difference to a calculated critical range.

$$Critical\ range\ = \sqrt{\chi^2_{\alpha}} \sqrt{\frac{p_j(1-p_j)}{n_j} + \frac{p_{j'}(1-p_j')}{n_j'}}$$

(Note: each pair will have a different critical range)

A particular pair of proportions is significantly different if:

$$\left|p_j - p_{j'}\right| > critical\ range\ for\ j\ and\ j'$$

Post – Hoc Analysis : Proportion Comparisons

Ex:-
Consider the above example.

We concluded that there were differences between the administrators, students and faculty using the chi-square test, but we still don't know which groups are different from each other. Now we will try to answer that question.

$$|p_A - p_S| = |0.63 - 0.4| = 0.23$$
$$|p_A - p_F| = |0.63 - 0.74| = 0.11$$
$$|p_F - p_S| = |0.74 - 0.4| = 0.34$$

$$\chi^2_{1\%,(2-1)\times(3-1)} = 9.2$$

Comparison	$\lvert p_j - p_{j'} \rvert$	Critical range	Result
Admin, Students	$0.63 - 0.4 = 0.23$	0.256	Proportions are not different
Admin, Faculty	$0.63 - 0.74 = 0.11$	0.238	Proportions are not different
Faculty, Students	$0.74 - 0.4 = 0.34$	0.282	Proportions are different

Chi Square Test for Independence

Consider the following hypothesis.

$$H_0: The\ two\ categorical\ variables\ are\ independent\ (i.e.\ no\ relationship\ between\ them)$$
$$H_1: The\ two\ categorical\ variables\ are\ dependent\ (i.e.\ there\ is\ a\ relationship\ between\ them)$$

The test-statistic will be the same but the degrees of freedom of the critical value changes,

$$\chi^2_{stat} = \sum_{all\ cells} \frac{(f_o - f_e)^2}{f_e} \sim \chi^2_{(r-1)\times(c-1)}$$

Where $f_e = \frac{row\ total * column\ total}{n}$

The rejection rule : if $\chi^2_{stat} > \chi^2_{\alpha,r-1\times c-1}$, reject H_0

Assumption: each cell in the contingency table has expected frequency of at least 1

Chi Square Test for Independence

Ex:-

The meal plan selected by 200 students is shown below:

	Meals per week			Total
Year of study	20	10	None	
1st year	24	32	14	70
2nd year	22	26	12	60
3rd year	10	14	6	30
4th year	14	16	10	40
Total	70	88	42	200

Is there a relationship between year of study and meal plan selected? Test at 5% level of significance.

Chi Square Test for Independence

Ex:-

$$H_0: Meal\ plan\ and\ year\ of\ study\ are\ independent$$
$$H_1: Meal\ plan\ and\ year\ of\ study\ are\ dependent$$

Expected frequencies can be calculated as:

	Meals per week			Total
Year of study	20	10	None	
1st year	70*70/200=24.5	88*70/200=30.8	42*70/200=14.7	70
2nd year	70*60/200=21	88*60/200=26.4	42*60/200=12.6	60
3rd year	70*30/200=10.5	88*30/200=13.2	42*30/200=6.3	30
4th year	70*40/200=14	88*40/200=17.6	42*40/200=8.4	40
Total	70	88	42	200

Chi Square Test for Independence

Ex:-

$$\chi^2_{stat} = \frac{(24-24.5)^2}{24.5} + \frac{(32-30.8)^2}{30.8} + \cdots + \frac{(10-8.4)^2}{8.4} = 0.709$$

$$\chi^2_{5\%,(4-1)\times(3-1)} = 12.592$$

Since $\chi^2_{stat} < \chi^2_{5\%,(4-1)\times(3-1)}$, do not reject H_0 at 5% significance.

Conclusion: There is an evidence to say that there is no relationship between meal plan and year of study at 5% level of significance.